

WDV Preview Template KR Reference

Framing and staging guide for ingesting the Standard Well Log Display Examples reference into the Knowledge Repository

Resource framing statement

This reference is a KR-controlled WDV preview-template and display-behavior resource. It should teach the KR how preview layouts are built from classified curve families. It must not be used as a formal mnemonic dictionary or as authority for vendor-specific tool identity.

1. Resource Identity

Field	Recommended value
Resource name	Standard Well Log Display Examples - WLV Reference
Resource type	WDV Preview Template / Display Behavior Reference
Domain	Well Log Viewer, Well Data Viewer, Knowledge Repository template layer
Primary ingestion target	KR Preview Template Layer; KR Display Behavior Layer; KR Track Object Model
Authority level	Internal WLV reference; candidate for controlled approval
Recommended initial status	Candidate - Requires Review Before Runtime Promotion
Primary runtime consumer	Backend WDV preview-template service and auto-build service
Non-runtime consumer	KR review UI / Knowledge Management page

2. Purpose and Scope

Purpose. Use this document to stage preview-template knowledge for the WDV. The KR should use it to decide which templates are eligible from available curve families, how tracks are assembled, where curves are placed, which renderer type is required, and what default scales should be offered.

Scope. This document covers preview-template structure, track positioning, display scales, curve-family grouping, track object types, and auto-build selection behavior. It does not define formal curve meanings. Curve identity remains governed by Energistics/PWLS, LAS/DLIS header evidence, vendor dictionaries, and approved KR curve-definition rules.

3. Trusted and Not Trusted Uses

Trusted for	Not trusted for
Preview template definitions; track positioning; display defaults; track object type selection; auto-build behavior; representative curve selection; WDV preview eligibility ranking.	Formal mnemonic definitions; vendor-specific tool identity; PWLS property-kind assignment; physical measurement definitions; regulatory reporting standards; overwriting source catalogue fields; automatic curve-family approval.

4. KR Tables Seeded by This Resource

KR table	Purpose	Key fields
kr_reference_sources	Registers this PDF/reference as a controlled KR source.	source_id, name, type, authority_level, version, file_hash, ingest_status, rule_status
kr_preview_templates	Defines WDV preview templates such as Triple Combo, Quad Combo, NMR, Cement Bond, Production Logging.	template_id, template_name, workflow_context, primary_usage, preview_eligible, auto_build_eligible, rule_status
kr_preview_template_tracks	Defines tracks within each template.	template_id, track_number, track_name, track_role, renderer_type, scale_behavior, display_notes
kr_template_curve_family_requirements	Defines required, preferred, optional, and fallback curve families for each template.	template_id, curve_family, requirement_level, preferred_mnemonics, max_auto_plotted_count, selection_priority
kr_template_scale_defaults	Stores scale defaults for preview plotting.	curve_family, unit_family, scale_type, scale_min, scale_max, display_direction, override_allowed, scale_confidence
kr_track_object_types	Defines supported WDV renderer/data object types.	object_type, compatible_data_types, default_renderer, preview_supported, auto_build_supported
kr_template_selection_rules	Ranks eligible preview templates from loaded curve families and workflow context.	rule_id, workflow_context, required_families, preferred_families, ranking_weight, action

5. Preview Template Library to Extract

Template	Workflow context	Core display content	Primary runtime use
Basic Lithology	Open hole / quicklook	Depth, GR/SP, caliper/bit size, formation tops	Quick lithology review and correlation
Open-Hole Triple Combo	Open hole	GR/SP/caliper + resistivity + density/neutron	Default formation-evaluation preview
Open-Hole Quad Combo	Open hole	Triple combo plus sonic	Expanded formation evaluation and seismic tie preparation
Reservoir Evaluation	Petrophysics	Triple combo + Vsh + PHIE/PHIT + Sw + pay flags	Petrophysical interpretation and pay review
Resistivity / Invasion Review	Open hole / saturation	GR plus shallow, medium, and deep resistivity	Invasion and saturation diagnosis
Density-Neutron Lithology	Open hole / lithology	GR + RHOB/NPHI + PEF + DRHO	Porosity, gas crossover, lithology discrimination
Sonic / Seismic Tie	Seismic tie / rock physics	GR + sonic + density + impedance + synthetic/reflectivity	Well-to-seismic calibration
Image Log Review	Borehole image	GR + borehole image + dip/fracture/facies tracks	Fracture, bedding, facies, borehole condition
NMR Evaluation	NMR / reservoir quality	NMR porosity, free/bound fluid, T2 distribution, permeability	Movable fluid and pore-system review
Cased-Hole Cement Bond	Cased hole	GR/CCL + CBL amplitude + VDL + bond index/cement map	Cement and zonal isolation evaluation
Production Logging	Cased hole / production	GR/CCL + spinner + pressure + temperature + holdup + completion schematic	Flow allocation and diagnostics
Well Integrity / Casing Inspection	Integrity	GR/CCL + caliper/radius/thickness + casing image	Corrosion, deformation, perforation damage
Core Integration	Core / facies	Logs + core photos + plug porosity/permeability + facies	Log-core calibration and facies validation
Completion / Intervention Review	Completion / workover	GR/CCL + schematic + perforations + pressure/temp/flow	Completion and intervention review

6. WDV Runtime Use

1. MDP loads selected curves, images, rasters, waveforms, or supporting objects to the WDV session.
2. Backend classifies loaded items into KR curve families, object types, and workflow contexts.
3. WDV asks the KR template service for eligible preview templates.
4. KR returns ranked templates based on required-family coverage, preferred-family coverage, workflow context, and renderer availability.
5. Backend assigns representative curves to each preview track; it does not plot every matching mnemonic.
6. Unused loaded curves remain available in the WDV left panel as loaded-but-unplotted data.
7. Backend returns a preview-layout contract to the frontend.
8. Frontend renders the backend-owned track layout and does not infer template eligibility or curve ranking locally.

7. Template Assembly Rules

Rule area	KR rule
Curve-family first	Identify curve families before plotting. Template selection should be based on families, not raw mnemonic counts.
Representative selection	For families with many curves, select representative curves by role and priority. Do not plot every matching curve automatically.
Resistivity	Classify by investigation depth: flushed/shallow, medium, deep/true. Prefer one or a small number from each role.
Porosity	Plot density and neutron together where available; sonic separately unless constrained.
Images/waveforms	Use image or waveform renderers, not ordinary line-curve renderers.
Interpretations	Place computed/interpreted products to the right of raw measurements. Visually distinguish from raw curves.

Flags/events	Render flags, tops, pay indicators, perforations, collars, and completion events as narrow categorical/event tracks.
Overrides	Allow per-template and project-level overrides of scale and track selection.

8. Track Object Types for WDV

Track object type	Use cases	Auto-build notes
line_curve	GR, RHOB, NPHI, DT, pressure, temperature	Allowed for confirmed scalar curves.
log_curve	Resistivity, permeability, mobility	Requires explicit logarithmic scale contract.
image_track	Borehole images, CGM/TIFF rasters, cement maps	Do not force into LAS curve model.
waveform_track	VDL, sonic waveform arrays	Needs waveform renderer; usually review required.
categorical_track	Lithology, facies, pay flags, QC flags	Narrow categorical column.
event_track	Tops, collars, perforations, sample points	Depth-aligned events; not continuous curves.
annotation_track	Text notes, shows, operational remarks	Text or interval annotation renderer.
schematic_track	Completion schematic, casing, tubing, packers	Mechanical/completion renderer.
computed_interpretation_track	Vsh, PHIE, Sw, net pay, impedance	Visually distinct from raw measurements.
tadpole_track	Dip angle and azimuth	Symbol/tadpole renderer, not line curve.
seismic_wiggle_track	Synthetic seismogram, VSP/corridor stack	Specialized seismic/wiggle renderer.

9. Display Scale Defaults and Confidence

Curve family	Default scale guidance	KR confidence treatment
Gamma Ray	Linear 0-150 API default; allow 0-200/custom.	High candidate when family/unit confirmed.
Resistivity	Logarithmic; support 0.2-20, 0.2-200, 0.2-2000 ohm-m.	High candidate for confirmed resistivity; depth role may need review.
Density	Linear 1.95-2.95 g/cc default; allow local range.	High candidate for RHOB/RHOZ with density units.
Neutron	Porosity display convention often 45 to -15 p.u. reversed, or 0.45 to -0.15.	High candidate for confirmed neutron porosity; store direction separately.
Sonic	DT commonly 40-140 us/ft; wider ranges for shear/Stoneley.	High/medium depending curve subtype.
PEF	Linear 0-10.	High candidate when PEF/PE family confirmed.
Porosity / saturation / Vsh	0-1 or 0-100%; preserve unit convention.	Medium/high only when computed family is clear.
Production / cased-hole / NMR / image / waveform	Use template defaults cautiously; many are context-specific.	Medium candidate or review required.

10. Ingestion, Review, and Promotion Workflow

Required control workflow
Observe -> Extract -> Suggest -> Review -> Approve -> Promote. Uploading or parsing this reference must not directly alter trusted KR runtime behavior.

1. Register the PDF/reference in kr_reference_sources with source metadata, file hash, authority level, and candidate status.
2. Extract candidate preview templates, track definitions, curve-family requirements, scale defaults, selection rules, and object types.
3. Store extracted items as candidate rules, preserving source references and extraction notes.
4. Review candidate rules in the KR review UI or controlled seed-review process.
5. Promote only approved rules into trusted runtime KR tables.
6. Expose approved templates through the backend WDV preview-template service.

11. Conflict Handling

- If this reference conflicts with Energistics/PWLS curve definitions, preserve PWLS/Energistics as the curve-definition authority and mark template/display conflict for review.
- If this reference conflicts with LAS/DLIS header evidence from a loaded file, preserve file evidence and mark preview assignment for review.
- If a scale default is incompatible with source units, do not convert silently; require unit normalization or review.
- If a template requires unsupported renderer capability, mark template preview unavailable or degraded rather than forcing a line-curve fallback.
- If a mnemonic appears in multiple contexts, select by curve family, unit, description, and workflow context rather than mnemonic alone.

12. Implementation Priorities

Priority	Implementation item	Reason
1	Seed Open-Hole Triple Combo and Quad Combo templates.	Highest WDV preview value and most common workflow.
2	Add family-based representative curve selection.	Prevents dumping all curves into preview tracks.
3	Add scale defaults for GR, resistivity, RHOB, NPHI, DT, PEF.	Provides immediate useful preview behavior.
4	Add non-line track object enum.	Prepares WDV for images, waveforms, categorical, event, and schematic tracks.
5	Add NMR, image, cement bond, production, integrity templates as capability-aware candidates.	Important but renderer/context dependent.

13. Core KR Instruction

Instruction to KR
Use this resource as: curve family + workflow context -> WDV preview template + track layout + renderer type + display defaults. Do not use it as: mnemonic -> formal curve definition.

14. Source Reference

Primary source: standard_well_log_display_examples.pdf, uploaded reference file. The source states its purpose is to summarize common well-log display templates, including display types, curves, positioning, scaling recommendations, and usage. It also states that exact curve names and scales vary by service company, vintage, units, basin, tool generation, and local interpretation practice.